JoLT: Joint Latent Trajectories for Context-Guided High-Resolution Tiled Generation

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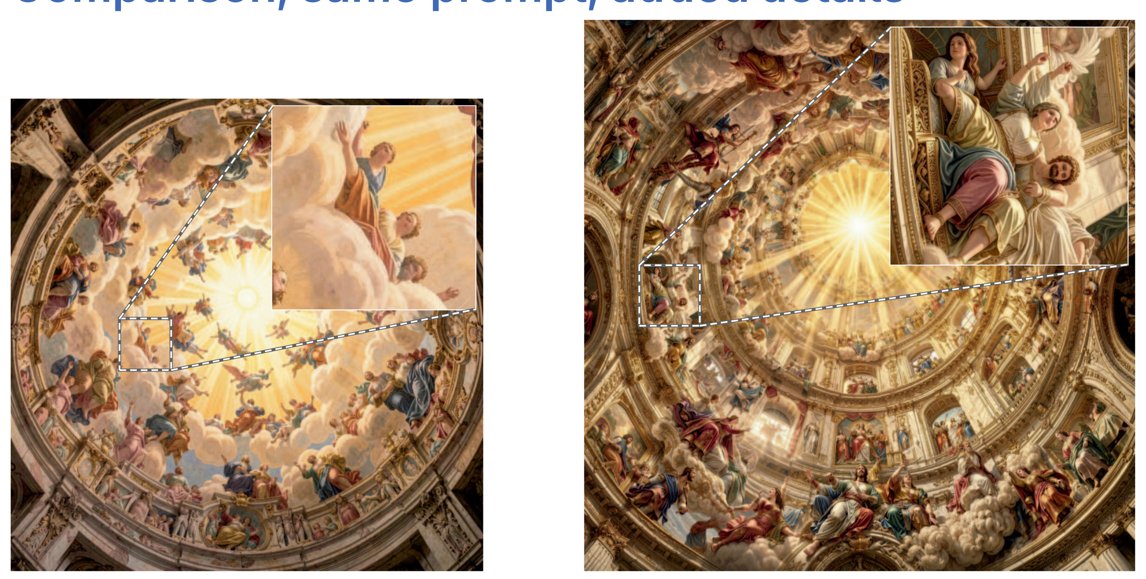
Project Page

1 Motivation & Task

- **Goal:** Generate globally coherent large-resolution images with dense local meaning.
- **Complex image:** high local semantic-information density within a globally understandable scene; texture, repetition, or sharpness alone do not count.
- **JoLT:** global structure and local detail shape each other throughout generation.
- **Controllability:** explicitly steer visual complexity, style, and local content without fine-tuning.

Comparison, same prompt, added details

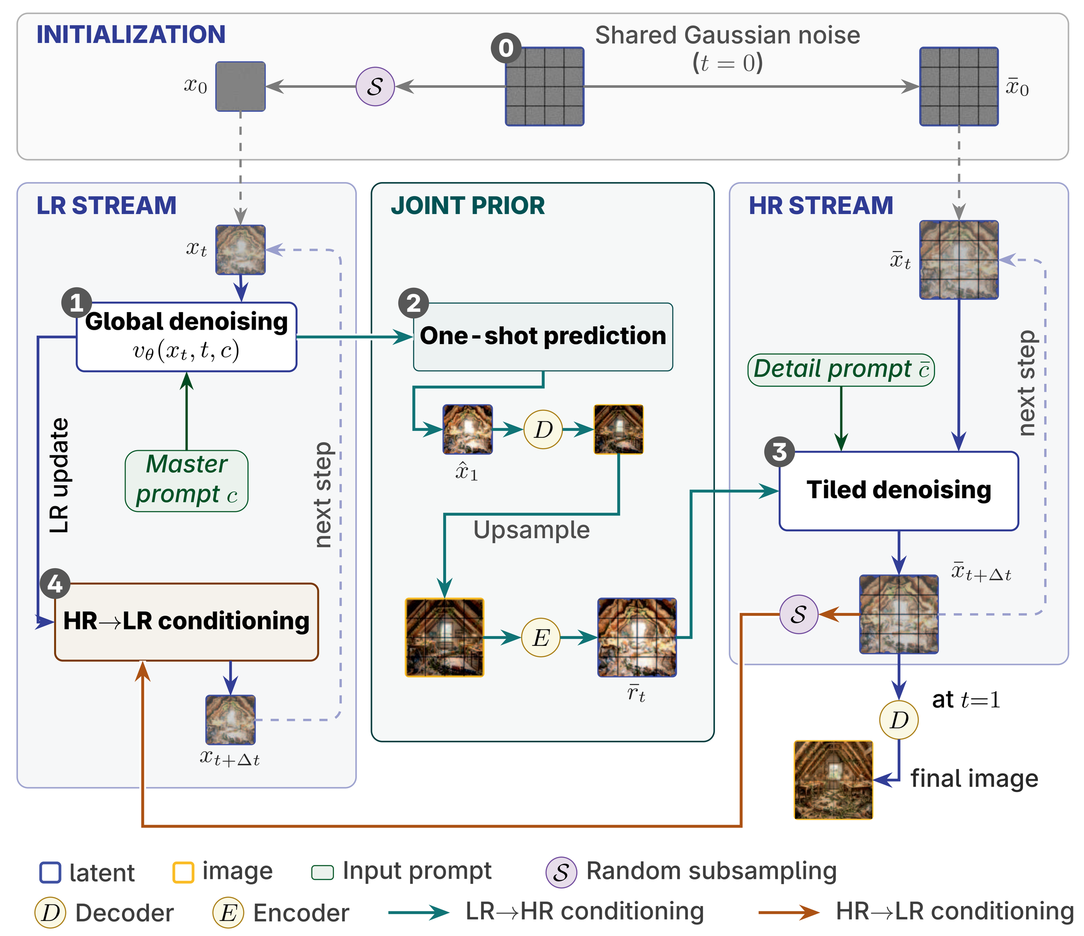


FLUX.2 Klein-9B 2048^2 | JoLT 4096^2

Our detail prompt adds coherent local structure at 4K res.

2 Method

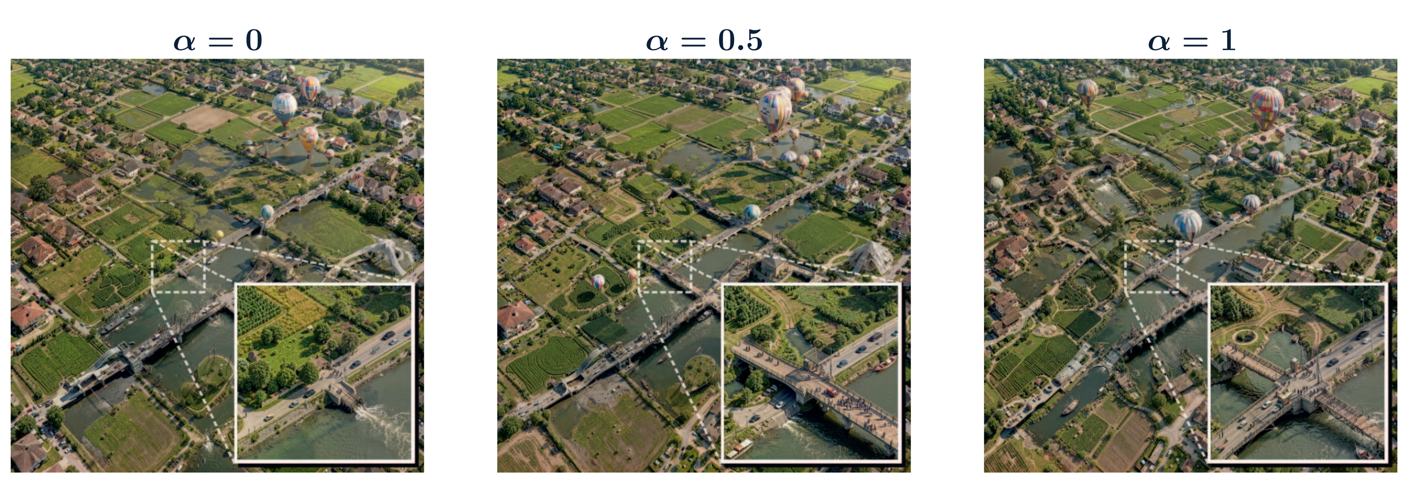
JoLT couples global low-resolution (LR) and tiled high-resolution (HR) trajectories. Prompt c sets the scene; detail prompt $\bar{c}$ controls added fine elements.



- **LR → HR:** the joint prior guides tiled layouts.
- **HR → LR:** we inject HR into LR latents by random subsampling, with controllable $\alpha \in [0, 1]$.

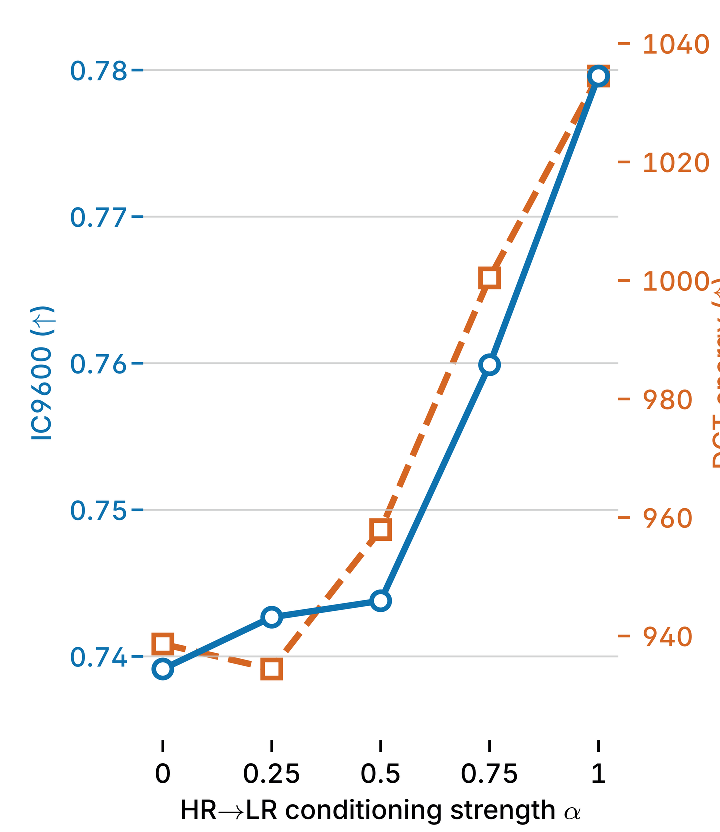
Cost: 127 s per 4K image on one H100 (four steps, BF16, no training). Baselines: 75–125 s.

3 α sweep: apparent complexity



Prompt: Aerial flooded suburb [...] Three α values; insets show local detail.

Complexity metrics. IC9600 estimates perceived complexity; discrete cosine transform (DCT) energy measures high-frequency structure. Higher means more complex, not better.



- **Feedback strength.** α controls how strongly HR feeds back into the LR trajectory; its effect depends on the detail prompt $\bar{c}$, not on α alone. At $\alpha = 0$, LR ignores HR, while HR still receives the LR prior.
- **In this sweep,** apparent complexity rises with α; prompt alignment stays stable.

4 Results

Same setup: all baselines are reimplemented on FLUX.2 Klein-9B at 4K with four steps and evaluated on 200 prompts.
Check the paper for quantitative metrics.

User preference for JoLT

vs.	More detailed	Better match	More artistic
SEGA	**77.3 ± 5.1%**	**69.0 ± 5.4%**	**68.0 ± 5.5%**
DemoFusion	**76.3 ± 5.1%**	48.5 ± 5.6%	46.3 ± 5.7%
ResDiT	**76.0 ± 5.1%**	**68.3 ± 5.5%**	**71.0 ± 5.4%**

2,700 pairwise judgments over 120 prompts.

Takeaways

- **Joint:** composition and detail develop together.
- **Bidirectional:** LR and HR guide each other at every step.
- **Control:** α sets feedback strength; $\bar{c}$ sets content and style.

5 Ablations

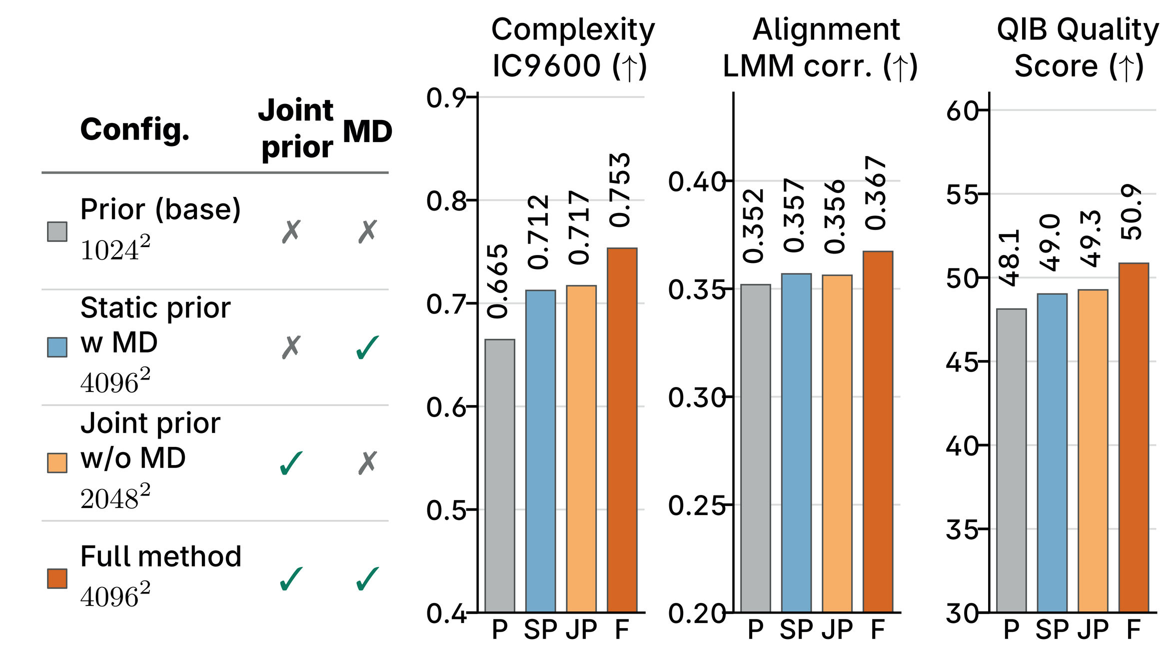
Detail prompt: what kind of detail

- **It can change:** Texture, lighting, material, style, details, or objects.
- **Varying only $\bar{c}$:** Prompt c, seed, and composition stay fixed.

"Enhance the image, add lots of details" | "pointillist style"

"Add iridescent caustic light reflections" | "Intricate grotesque imagery"

Which components matter?



Four configurations isolate the LR prior, joint LR–HR denoising, and MultiDiffusion (MD); 200 prompts.

- **Joint vs. static, 4K:** IC9600 +5.7%; prompt alignment +2.9%; quality +3.7%.
- **With MD, 4K:** IC9600 +5.1%; prompt alignment +3.1%; quality +3.2%.

Metrics. LMM4LMM correspondence = prompt alignment. QIB = Qwen-Image-Bench (Qwen3.6-27B) quality: realism, detail, resolution.